JOÃO PAULO NOGUEIRA

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No visa sponsorship required (eligible for the UK High Potential Individual visa)

EDUCATION

Institut Polytechnique de Paris *Sep. 2023 – Jul. 2025*
MSc in Computer Science (Computer Science for Networks) *Paris, France*
Mention Très Bien (Highest Honors); thesis at Nokia Bell Labs

- **Coursework**: Machine Learning; Graph Learning; Optimization; Information Theory; Formal Verification.

Polytechnic Institute of Porto *Sep. 2020 – Jul. 2023*
BSc in Electrical and Computer Engineering *Porto, Portugal*
ECTS grade A, Highest Honors

- **Coursework**: Applied Artificial Intelligence; Algorithms and Programming; Linear Algebra; Statistics.

PUBLICATIONS

- W. Sun, **J. P. Nogueira**, D. Verchere, M. Acher, A. Silva. "Externalized CPDAG Summaries Improve LLM Causal Deduction." **NeurIPS 2026** (poster). **arXiv:2609.31071**.
- **J. P. Nogueira**, W. Sun, A. Silva, L. Zumot. "Certainty-Guided Reasoning in Large Language Models: A Dynamic Thinking Budget Approach." Preprint, 2025. **arXiv:2509.07820**.

RESEARCH & PROFESSIONAL EXPERIENCE

Machine Learning Research Intern Apr 2025 – Sep 2025
Nokia Bell Labs, Machine Learning and Systems Department *Paris, France*

- Built and maintained the project's research code base in Python (data collection, analysis, visualisation) feeding multi-day inference and evaluation runs on H100 GPUs.
- Implemented a custom vLLM logits processor to intervene in inference (budget forcing), and evaluated parallelization and data-management frameworks to resolve pipeline bottlenecks.
- Proposed Certainty-Guided Reasoning, an adaptive thinking-budget method for reasoning LLMs that maintained accuracy on AIME2024/2025 while eliminating millions of tokens in aggregate.
- Selected as a Top 4 AI finalist in the Nokia Bell Labs Global Student Program from a worldwide pool of interns, the first France-based intern to reach that stage. Received its Outstanding Innovation Award.
- Designed the *Grade* evaluation metric, which penalizes confident errors and rewards calibrated abstention.
- Work selected for an oral presentation at the LINCS Annual Workshop, delivered by my supervisor.
- Co-authored a NeurIPS 2026 paper showing that externalizing a schema-constrained CPDAG summary before answering raises Qwen3.5-27B from 73.0 to 86.4 F1 on Corr2Cause.

Industrial Research Project – Networks Nov 2023 – Jun 2024
Institut Polytechnique de Paris *Paris, France*

- Implemented a cognitive-radio layer over LoRa: spectrum sensing and dynamic channel switching.
- Developed the MATLAB simulation, then validated sensing and channel switching on SDR hardware.
- Cassiopée Award – Best Student Project 2024 (top among 60+ projects).

Robotics Research Intern Feb 2023 – Jun 2023
INESC TEC, Robotics and IoT Laboratory, University of Porto *Porto, Portugal*

- Wrote a low-latency C driver on the RP2040 for an LD19 LiDAR (raw packet parsing, point-cloud output).
- Built a UART pipeline streaming scans to host devices, now used to support navigation on the lab's agricultural field robots; extended an Android (Java) app to map the environment in real time.

SELECTED PROJECTS

Learning to Walk: RL Locomotion on a Unitree Go2

- One-week team project supervised by Prof. Antoine Cully at the Imperial College Robotics Summer School (July 2026): trained PPO locomotion policies in massively parallel Isaac Gym.
- Tuned reward terms and hyperparameters in the walk-these-ways framework; the resulting policy ran zero-shot on the physical robot, onboard with no fine-tuning.

Sampling-Based Motion Planning and SLAM

- Benchmarked RRT and RRT* on path quality against computation time; solved narrow passages by biasing samples towards obstacles, finding that a 20–40% bias maximizes success rate.
- Implemented ICP scan registration with a simplified GraphSLAM back-end for 2D LiDAR mapping.

SMT(LRA) Constraint Learning

- Reimplemented the INCAL algorithm (Kolb et al., IJCAI 2018) in PySMT: an SMT solver learns interpretable constraints over Boolean and linear real-arithmetic terms that separate feasible from infeasible examples.

Real-Time Facial Recognition

- Fine-tuned YOLOv8m on a self-collected, self-labelled dataset for closed-set identification; ran live inference with a confidence threshold, evaluated on precision / recall / F1.

Real-Time Train Crowding Forecasting (SNCF-Transilien)

- Benchmarked linear and polynomial regression, an MLP, and Random Forest on noisy, incomplete sensor data; engineered aggregated lag features to reach RMSE 0.0229 against a 0.158 baseline.

Machine Learning for Mutation Testing

- Clustered mutants of an FSM flight-booking system by structural features (K-Means, DBSCAN, HDBSCAN), then verified equivalence within clusters with FSM-derived test suites to prune the mutant set.

Embedded Tetris Table

- Wrote Tetris in C on an ATmega328P with 2 KB of RAM (bitwise collision grid), driving 300 WS2812B LEDs with a state machine that protects their 800 kHz timing from Bluetooth UART interrupts.

AWARDS & DISTINCTIONS

- Outstanding Innovation Award and Top 4 AI finalist, Nokia Bell Labs Global Student Program (2025)
- Offered a funded CIFRE PhD by Nokia Bell Labs and Institut Polytechnique de Paris (2025)
- Mention Très Bien (Highest Honors), MSc in Computer Science, Institut Polytechnique de Paris (2025)
- Cassiopée Award, Best Student Project, Institut Polytechnique de Paris (2024)
- Highest Honors, BSc; ranked 1st of 19 in Applied AI, Polytechnic Institute of Porto (2023)

SKILLS

Programming	Python; C / C++; Java; MATLAB; SQL
ML	PyTorch; vLLM; scikit-learn; Isaac Gym; Weights & Biases; NumPy / Pandas / Polars
Systems	Linux; Git; Docker; Kubernetes; H100 GPUs; Databases; LaTeX
Languages	Portuguese (native); English (native); French (B2); Spanish (B1); Italian (A1)
Interests	Chess; swimming; hiking; strategy and open-world games (Civilization VI, Skyrim)